Line (GAC): A ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING

Competency: A5 Describe basic road and highway profiles

Objectives

To be competent in this area, the individual must be able to:

- Understand the categories of road construction projects.
- Understand basic specifications for roads as defined by MOT, MMCD and local government.

LEARNING TASKS	CONTENT
1. Review categories of projects	• Commercial • Industrial • Residential • Provincial
2. Describe types of owners	• Provincial • Federal • Municipal • Private
3. Describe common types of asphalt work	• Open ditch road/shoulder • Sidewalk • Curb and gutter (contained) • Crowns • Graders • Runoff • Super elevations
4. Review specifications for different types of roads	• MOT, MMCD, Local Government, Architectural (building specific)
5. Describe cross section of road	• Elements to consider ○ Civil work ○ Prep for paving

Line (GAC): A ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING

Competency: A6 Describe future paving/construction trends

Objectives

To be competent in this area, the individual must be able to understand the paving trends for the future.

LEARNING TASKS	CONTENT
1. Describe green roads	• Pave stones • Porous asphalt • Recycling
2. Describe new paving materials	• Epoxy, coloured, modified asphalt
3. Describe new sub-bases	• Pumice
4. Describe floating roads	• Styrofoam
5. Describe added value enhancements	• Street print

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B1 Safety orientation

Objectives

To be competent in this area, the individual must be able to understand and demonstrate safe working practices.

LEARNING TASKS	CONTENT
1. Describe PPE	• Workers are responsible to provide general purpose gloves, safety footwear, hard hat • Employers are responsible for providing all other safety items • Employers must ensure that PPE is properly worn when required, and is cleaned, inspected, maintained • Employers must ensure that workers are adequately trained: correct use, limitations, maintenance
2. Describe hazard assessment	• Chemical: inhalation (e.g. carbon monoxide, hydrogen sulphide, welding fumes, asbestos), ingestion, absorption, injection (i.e. high pressure fluids) • Sensory: noise, vibration, heat, cold • Environmental: bacteria, micro organisms, viruses • Physical: repetitive stain, strains and sprains, back injuries, trauma, etc.
3. Describe key elements of safety policy	• Safety orientation (company manuals)
4. Describe required certifications	• WHMIS, First Aid
5. Describe safe rigging practices	• WorkSafeBC rigging requirements (Part 15)
6. Describe industry/site and equipment specific risks	• The most common cause of injury (slips and falls) • Hearing loss: gradual exposure (long term) or traumatic injury (explosion, impact noises) crushing/ impact • Underground and overhead services and hazards (BC hydro 7 steps, BC 1Call) • Motor vehicle accidents are common in the workplace • Burns from hot substances or objects

LEARNING TASKS	CONTENT
	• Batteries, tires, pressurized hydraulic and fuel systems • Inexperience and lack of training – working beyond your limits • Complacency, boredom, inattention, lack of focus on safety • Drugs and alcohol
7. Describe how to secure site and equipment for public safety	• Vandal proof take ancillary parts off • Park safely block elevated equipment, use parking brakes and chocks

Line (GAC) B ASPHALT PAVING SAFETY

Competency: B2 Describe working safely around haul trucks

Objectives:

To be competent in this area, the individual must be able to understand and demonstrate safe working practices while working around haul trucks.

LEARNING TASKS	CONTENT
1 Describe consequences of lapses in communication	• Maintain eye contact
2 Identify equipment specific hazards	• Pinch points • Blind spots • Machine cleanliness

Achievement Criteria

Performance — Learner will demonstrate safe working practices while working with and around haul trucks.

Conditions — This competency will be measured both in technical training and within the workplace. Learners will use suitable communication skills with the haul truck operator.

- direct eye contact
- appropriate hand signals
- radio communication

The learner will demonstrate awareness general safety issues while working with or near halt trucks

- blind spots,
- pinch points
- turning radius
- overhead clearances.

Criteria — Ground workers will position themselves so that they remain in a safe location during the process of directing the truck to the work area.

Paver operators will ensure that the area between the haul truck and the paver remain clear of workers and/or equipment.

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B3 Describe safety working around and operating power brooms

Objectives

To be competent in this area, the individual must be able to understand and demonstrate safe working practices while working around power brooms.

LEARNING TASKS	CONTENT
1. Describe effects of noise and dust	• Use proper PPE including Hi-Vis and dust mask
2. Describe dangers associated with vehicle movement	• Power brooms move quickly, collision hazard
3. Describe dangers associated with moving parts	• Spinning brushes and attachments can cause injury

Achievement Criteria

Performance	Learner will demonstrate safe working practices while working with and around power brooms. This competency will be measured both in technical training and within the workplace.
Conditions	The learner will use proper PPE and position themselves in a safe area during sweeping procedures.
Criteria	The learner will consistently demonstrate the correct use of PPE's and choice of safe location during sweeping procedures.

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B4 Describe safety working around and operating air hammers

Objectives

To be competent in this area, the individual must be able to understand and demonstrate safe working practices while working with air hammers.

LEARNING TASKS	CONTENT
1. Describe noise and vibration hazards	• Use proper PPE including hearing protection
2. Describe hazards of flying debris	• Keep clear to avoid being hit by flying debris
3. Describe ergonomics and safe equipment handling	• Air hammers are heavy, care must be taken to avoid strain while operating or stowing

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B5 Describe safety working around and operating propane tanks

Objectives

To be competent in this area, the individual must be able to understand and demonstrate safe working practices while operating propane tanks.

LEARNING TASKS	CONTENT
1. Identify potential fire hazards while working around propane tanks	• Discuss properties of compressed gas including explosion/fire
2. Demonstrate proper storage and handling	• Secure for transport • Stow attachments e.g., Tiger Torch
3. Demonstrate proper connection procedures and check for leaks	• Use of soapy solution to check for leaks • Proper connection of threaded fittings

Achievement Criteria

Performance	Learner will demonstrate safe working practices while working with propane bottles and attachments. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate correct connection procedures (bottle to attachment), check for leaks and demonstrate correct securing procedures for stowing and transport of bottles and attachments.
Criteria	The learner will demonstrate the proper connection, securement and storage procedures. The learner will check for leaks at fittings.

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B6 Describe weather effects and precautions

Objectives

To be competent in this area, the individual must be able to understand the effects of weather with regard to safety and will know how to take safety precautions to protect ones self.

LEARNING TASKS	CONTENT
1. Describe the effects of temperature on the body	• Excess heat can cause exhaustion • Radiant heat from pavement
2. Describe the importance of proper hydration	• Drink plenty of fluids, especially during asphalt laydown • Recognize signs and treatment of heat exhaustion and heat stroke
3. Define effects of exposure to direct sun	• Risk of skin cancer • Use proper protective clothing, hats, etc • Use sunscreen

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B7 Describe safety and traffic control procedures

Objectives

To be competent in this area, the individual must be able to understand traffic control signage and safety procedures.

LEARNING TASKS	CONTENT
1. Describe traffic control procedures (TCP) Certification	• Certification process Radiant heat from pavement
2. Review TCP training content	• Summary of TCP course content
3. Review TCP regulations	• Number of TCPs required • Signage • Signals • Communication

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B8 Describe plant site and gravel pit safety

Objectives

To be competent in this area, the individual must be able to understand the hazards associated to an asphalt plant and gravel pit.

LEARNING TASKS	CONTENT
1. Define noise hazards and precautions	• Hearing protection
2. Describe site specific hazards	• Crushing • Moving equipment • Flying debris • Dust

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B9 Describe light vehicle transporting procedures

Objectives

To be competent in this area, the individual must be able to understand general company and provincial rules regarding vehicular transportation.

LEARNING TASKS	CONTENT
1. Describe procedures for transportation of vehicles under 5000 kg	• Follow company safety rules
2. Describe procedures for transportation of vehicles over 5000 kg	• Transport Canada rules apply

Line (GAC): B ASPHALT PAVING SAFETY

Competency: B10 Complete fatigue awareness training

Objectives

To be competent in this area, the individual must be able to understand and complete fatigue awareness training.

LEARNING TASKS	CONTENT
1. Define mental fatigue	• Combination of cumulative sleep debt and the time of day circadian rhythms
2. Define a work schedule that helps reduce fatigue	• Work schedules need to be designed properly to reduce the risk of creating cumulative or acute sleep debt
3. Discuss available training	• Examine sources of fatigue awareness training e.g., www.Shiftwork.ca

Line (GAC): C ASPHALT PRODUCTION AND PRODUCTS

Competency: C1 Describe crushing process and paving aggregate production

Objectives

To be competent in this area, the individual must be able to understand the processes of crushing and aggregate plants.

LEARNING TASKS	CONTENT
1. Describe gravel pits, quarries	• Types of pits
2. Describe crushing equipment	• Jaw, cone, impact
3. Describe the screen	• Description and purpose
4. Describe conveyors	• Types of conveyors
5. Describe feeders	• Description and purpose
6. Explain stockpiling and handling	• How materials is stored prior to use
7. Describe washing procedures	• Methods and purpose
8. Describe types of aggregate transportation	• Barge and trucking and rail cars

Line (GAC): C ASPHALT PRODUCTION AND PRODUCTS

Competency: C2 Describe types and specifications of aggregates

Objectives

To be competent in this area, the individual must be able to understand the basic characteristics of asphalt construction products.

LEARNING TASKS	CONTENT
1. Describe aggregate processing	• Screening • Crushing • Washing
2. Describe characteristics	• Permeability • Load bearing • Resistance to shearing • Gradation • Plastic limit • Liquid limit
3. Describe common products	• Describe characteristics and most common uses for each ○ Pit run ○ Screened road base 3” minus ○ Bedding sand ○ Crushed road mulch ○ Asphalt aggregates ○ Drain rock ○ Chips, driveway chips ○ Recyled asphalt ○ Concrete sand, C33 ○ Stucco sand

Line (GAC): C ASPHALT PRODUCTION AND PRODUCTS

Competency: C3 Describe asphalt plant operation

Objectives

To be competent in this area, the individual must be able to understand the basic processing functions between a drum plant and a batch plant.

LEARNING TASKS	CONTENT
1. Describe the operation of a drum plant	• Cold feeds • Conveyor systems • Mixing process • Storage silo • Load out
2. Describe the operation of a batch plant	• Cold feeds • Hot stone elevator • Screen deck • Hot bins • Weigh bins • Pug mill • Slat conveyor

Line (GAC): C ASPHALT PRODUCTION AND PRODUCTS

Competency: C4 Describe asphalt products and uses

Objectives

To be competent in this area, the individual must be able to understand where and how to find product information as described in the appropriate ministry regulations.

LEARNING TASKS	CONTENT
1. Describe the key elements of the MMCD	• Master Municipal Contract Documents
2. Describe the role of the MOT	• Ministry of Transport • Quality control • Quality assurance • Quality audit
3. Describe the role of the DOT	• Department of Transport • Design build standards for highways contruction
4. Describe Engineer Specs	• Apply to private property
5. Describe role of Architect	• Variety of publications
6. Describe other asphalt and paving materials	• Paving blocks/stones • Green roads • Concrete

Line (GAC): D ASPHALT ENVIRONMENTAL CONSIDERATIONS

Competency: D1 Describe environmentally friendly properties

Objectives

To be competent in this area, the individual must be able to understand the environmentally friendly properties of asphalt.

LEARNING TASKS	CONTENT
1. Define Green properties of asphalt	• Not toxic when laid • Does not leach • Can be recycled • Can be used with other materials and used again: RAP [Recycled Asphalt Products]

Line (GAC): D ASPHALT ENVIRONMENTAL CONSIDERATIONS

Competency: D2 Describe use of storage areas

Objectives

To be competent in this area, the individual must be able to understand the appropriate storage considerations for asphalt products.

LEARNING TASKS	CONTENT
1. Describe Type 1 storage	• Spill contained areas – dangerous goods (tank and containment pool) • Fuels (diesel bunker) • Waste • Primer and SS1 • Natural gas • Asphalt oils
2. Describe Type 2 storage	• Self contained • Liquid asphalt • Fuels (bio diesel)
3. Describe use of recycled materials in hot mix asphalt	• Cost savings • Mobile recycling
4. Describe use of recycled materials in asphalt plant burners	• Temperature • Handling precautions

Line (GAC): D ASPHALT ENVIRONMENTAL CONSIDERATIONS

Competency: D3 Describe asphalt plant and placement emissions

Objectives

To be competent in this area, the individual must be able to understand emissions regulations and control systems for asphalt plants and placement.

LEARNING TASKS	CONTENT
1. Discuss role of Metro Vancouver (formerly GVRD) in enforcement and regulation of emissions	• 21 communities
2. Describe role of MOE in establishing regulations	• Provincial regulations
3. Describe use of Bag house	• Dust collection technologies
4. Describe emission controls	• Heating asphalt

Line (GAC): E ASPHALT TESTS AND QUALITY CONTROL

Competency: E1 Determine asphalt mix design

Objectives

To be competent in this area, the individual must be able to understand mix design.

LEARNING TASKS	CONTENT
1. Describe asphalt/aggregate composition	• Marshall method • Super pave method
2. Describe types of mix designs	• Mix
3. Describe mix temperature characteristics	• Hot mix asphalt HMA – upper and lower • Warm mix asphalt (WMA) • Cold mix aspalt

Line (GAC): E ASPHALT TESTS AND QUALITY CONTROL

Competency: E2 Confirm that samples match mix design

Objectives

To be competent in this area, the individual must be able to understand how to confirm and match mix design at plant.

LEARNING TASKS	CONTENT
1. Describe aggregate sampling	• Weigh bins
2. Describe belt sampling	• Drum mixer
3. Describe lab tests	• Sample of mix to lab
4. Describe the process of taking mix temperature readings	• Temperature measurement tools and recording equipment

Line (GAC): E ASPHALT TESTS AND QUALITY CONTROL

Competency: E3 Determine haul truck samples

Objectives

To be competent in this area, the individual must be able to understand how to confirm and match mix design at truck.

LEARNING TASKS	CONTENT
1. Describe mix sampling	• Confirmation ○ Of an asphalt sample ○ Temperature test ○ Extraction/gradation test
2. Describe quality control	• Requirement: truck sample randomly taken at job site (by testing agency or company of certified)
3. Describe how to bag samples	• For backup

Achievement Criteria

Performance	Learner will demonstrate how to accurately take a representative sample at the jobsite. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate correct sampling procedures.
Criteria	The learner will demonstrate the ability to take a representative samples at the jobsite using various sampling locations. The learner will demonstrate correct bagging procedures.

Line (GAC): E ASPHALT TESTS AND QUALITY CONTROL

Competency: E4 Determine compaction levels

Objectives

To be competent in this area, the individual must be able to understand two methods to determine compaction levels.

LEARNING TASKS	CONTENT
1. Describe two methods to determine compaction	• Nuclear densometer • In situ coring sample

Achievement Criteria

Performance	Learner will achieve a minimum grade of 70% on a quiz including questions covering content from each learning task.
Conditions	N/A
Criteria	N/A

Line (GAC): F ROAD STRUCTURES

Competency: F1 Describe drainage, grade and base

Objectives

To be competent in this area, the individual must be able to understand basic pre-asphalt road construction.

LEARNING TASKS	CONTENT
1. Describe proper road drainage	• Environmental considerations
2. Describe sub grade	• Existing soil conditions • Geo-textiles
3. Describe sub-base	• Aggregation and alternative materials
4. Describe base	• Virgin product • Quary, pit or dredge or river extraction/material • Recycled products